

LAKSHYA JINDAL

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EDUCATION

International Institute of Information Technology, Hyderabad

Aug 2023 – Present

B.Tech + MS(By Research) in Electronics and Communication Engineering

Hyderabad, India

Hiranandani Foundation School Thane

June 2021 – June 2023

ISC, PCM and Computer Applications - 94.25%

Mumbai, India

EXPERIENCE

Undergraduate Researcher | Robotics Research Centre, IIIT Hyderabad

April 2025 – Present

Research Assistant

On-site

- Designed and implemented scalable motion planning algorithms in C++ and Python, improving performance through iterative optimization, testing, and debugging.
- Built modular software pipelines for large-scale algorithm evaluation, improving reproducibility, reliability, and maintainability.
- Collaborated with researchers to develop production-quality software from research prototypes, participating across design, implementation, evaluation, and continuous iteration.

PROJECTS

Audio Amplifier Design | *Analog Electronics, LTSpice, Circuit Design, Signal Processing* | [GitHub](#)

- Designed and built a **multi-stage audio amplifier** capable of delivering **more than 1.5W output power** to a 10 Ω load using a $\pm 5V$ supply. Implemented stages including **differential pre-amplifier**, **common-emitter gain stage**, **band-pass filter (20 Hz–20 kHz)**, and **Class-AB power amplifier** for low distortion and high efficiency.
- Simulated and validated the design in **LTSpice** and hardware, achieving accurate gain, stable frequency response, and clear audio reproduction through oscilloscope-based testing.

Pulse Oximeter Design | *Analog Electronics, Arduino, LTSpice, Biomedical Instrumentation* | [Reference](#)

- Designed and built a **transimpedance amplifier-based pulse oximeter** to measure **blood oxygen saturation (SpO₂)** using synchronized **red and infrared LEDs** with a photodiode for optical signal acquisition.
- Developed the analog front-end comprising a **transimpedance amplifier, signal conditioning, and synchronized LED driving circuitry**, while using an **Arduino** to process sensor data, compute SpO₂ values, and display real-time readings. Modified the reference design by implementing a **battery-powered** system instead of the original solar-powered architecture.

4-bit Carry Lookahead Adder (CLA) | *VLSI Design, Verilog, CMOS, MAGIC, FPGA* | [GitHub](#)

- Designed and implemented a **4-bit Carry Lookahead Adder (CLA)** across multiple abstraction levels, including **HDL, transistor-level design, layout (MAGIC), and FPGA implementation** using **TSMC 180nm technology**.
- Utilized **static CMOS** and **TSPC logic** to optimize speed and area, and verified functionality through simulation and hardware testing.

RISC-V Processor Design | *SystemVerilog, Computer Architecture* | [GitHub](#)

- Designed sequential and 5-stage pipelined RISC-V processors implementing hazard detection, forwarding, and control logic to improve execution efficiency.
- Verified processor correctness through simulation and comprehensive testing across control hazards, data hazards, and instruction execution scenarios.

TECHNICAL SKILLS

Languages: C, C++, Python, Verilog HDL, SystemVerilog, SQL, Shell, SPICE

Hardware Design: Analog Circuit Design, Digital Design, CMOS, Transistor-Level Design, FPGA, PCB Prototyping

EDA & Tools: LTSpice, MATLAB, MAGIC, NGSpice, Arduino IDE, Git, Linux, ROS

ACHIEVEMENTS/EXTRACURRICULARS

- Achieved a peak rating of **1269** on **Codeforces**
- Secured **3rd Place in Megathon 2025**, Built Multi-Modal AI Solution for Estimating Damage Claim for Cars.
- Solved 200+ Data Structures and Algorithms problems across Codeforces, LeetCode and CodeChef.
- Secured **AIR 4391** in JEE Mains 2023 and **AIR 10931** in JEE Advanced 2023
- Cleared **KVPY 2022** and **Pre-RMO 2019**

RELEVANT COURSEWORK

Data Structures and Algorithms | Networks Systems and Signals | VLSI Design | Probability & Random

Processes | Signal Processing | Robotics: Planning and Navigation | Intro to Processor Architecture |

| Computer Vision | Statistical Methods in AI | Communication Theory | Mechatronics System Design |

| Introduction to IoT | Analog Electronic Circuits | Mobile Robotics | Data Analytics I | Digital VLSI Design |